

# Experiments: Part II

## PS200C Quant Methods in Politics, Spring 2026

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# Where we are

- Part I: random assignment makes the simple difference in means an unbiased estimator of the ATE.
- Part II: Be better than **unbiased**, be closer to correct in your sample, using design and analysis levers.
- Roadmap:
  - 1 Why unbiasedness isn't the whole story
  - 2 Covariate adjustment (precision, not bias)
  - 3 Blocking (better-than-random balance by design)
  - 4 Non-independence: dependent residuals and clustered assignment
  - 5 Wrap and takeaways

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- What we actually care about is how far you are from the truth in your experiment.
- As a property of a procedure, we look at **root mean squared error**:

$$\text{RMSE}(\hat{\tau}) = \sqrt{\mathbb{E}[(\hat{\tau} - \tau)^2]} = \sqrt{\text{Var}(\hat{\tau}) + \text{Bias}(\hat{\tau})^2}$$

- For an unbiased estimator,  $\text{RMSE} = \sqrt{\text{Var}\hat{\tau}} = SD$ , so the goal is to “minimize variance”.

# Two different “variances”

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We care about both but for different reasons.

# Running example: a GOTV experiment

- **Unit:** a registered voter.
- **Treatment  $D$ :** a get-out-the-vote contact (mailer, canvass, etc.) vs. no contact.
- **Outcome  $Y$ :** did they vote? (0/1)
- **Covariates  $X$ :**
  - `past_turnout`: an index of voting in the last several elections. Strong predictor of  $Y$ . (This is a pre-treatment proxy for  $Y(0)$ .)
  - `age`, `partisan strength`: weaker predictors.
- **Structure:** voters are nested in households (4 per household) and households in precincts (10 per precinct). 100 precincts,  $N = 4000$ .
- **True effect:**  $\tau = +0.05$  on the probability of voting.
- **We will estimate  $\hat{\tau}$  many times under different designs and analyses**, and compare the resulting distributions.

# What does a single experiment look like? Check balance.

We randomly assign half the sample to treatment. Then look at the covariate means by treatment status (a “balance table”):

| Variable     | Treated | Control | Difference |
|--------------|---------|---------|------------|
| past_turnout | -0.028  | -0.001  | -0.027     |
| age          | -0.015  | -0.016  | +0.002     |
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Imbalance on (prognostic) covariates surface reasons to worry about (and opportunities to reduce) this gap.

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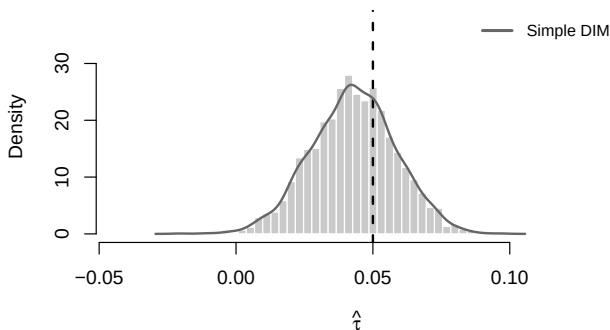
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- Logit/probit are fine too — but their coefficients are *not* the ATE; require computing marginal effects to recover the ATE.
- So: Don't let the binary outcome scare you away from OLS.

# Simulating the simple design

2000 replications. Each one: assign  $D$ , draw  $Y$ , run  $\text{lm}(Y \sim D)$ .

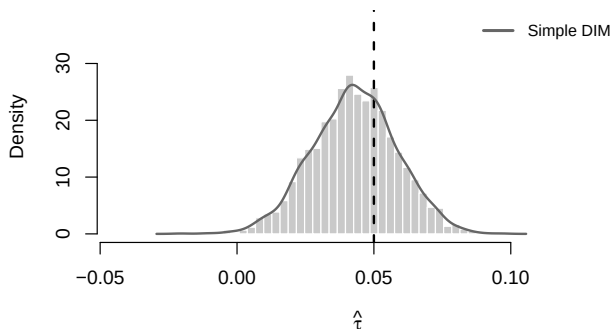
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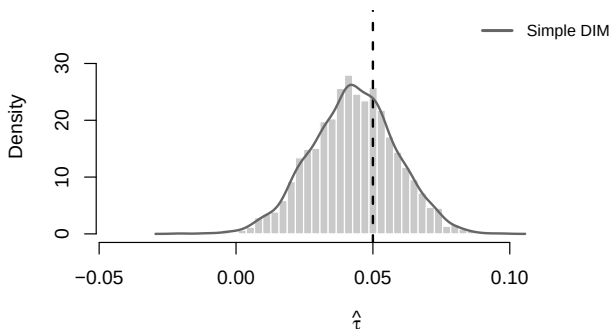


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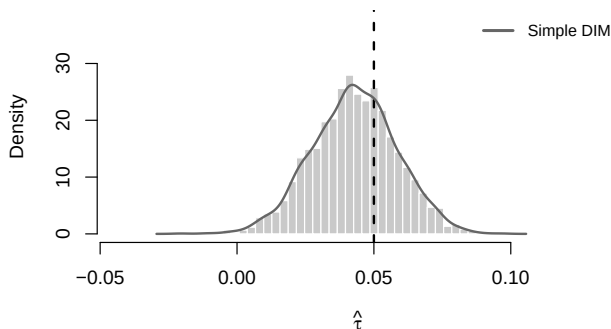


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We have real uncertainty, but the estimated SE knows it.

# Covariate adjustment: motivation

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- Thus we are adjusting to reduce variance (gaps) *not* bias.
- Strong predictors (`past_turnout`) help a lot. Weak ones (e.g. `partisan strength`) barely help, but if you have enough sample, okay.

# Adjustment “perfects” the balance the regression sees

Why does OLS with  $X$  work?

- One way to think of it: OLS splits  $D$  into the part predicted by  $X$  and the part orthogonal to  $X$ , and uses only the orthogonal part to estimate  $\tau$ .

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- So whatever variance was driven by that imbalance also goes away.
- (Formal version — the Frisch-Waugh-Lovell theorem — in the appendix.)

# Better still: The Lin Estimator

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Lin (2013) showed a small modification is better:

$$Y_i = \alpha + \tau_{Lin} D_i + \beta'_1 (X_i - \bar{X}) + \beta'_2 D_i \cdot (X_i - \bar{X}) + \varepsilon_i$$

- Demean  $X$ , then interact with  $D$ .
- $\tau_{Lin}$  is interpretable directly as an ATE.
- Removes a small ( $O(1/N)$ ) bias from plain regression adjustment that Freedman (2008) flagged; deals with "weird weights under heterogenous effects" we will get to later.
- Use as your default, pre-specify the covariates.

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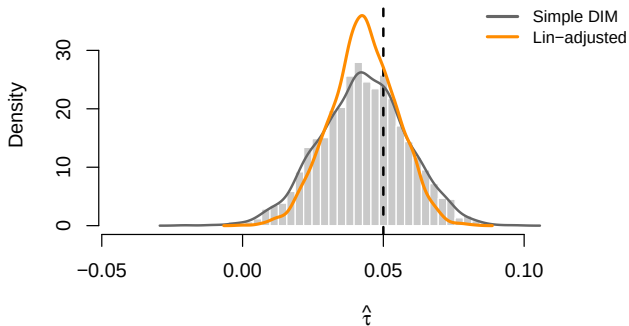
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(See appendix for ample details)

## Simulation: + Lin-adjusted

## Add regression adjustment (Lin)



- The orange curve is the Lin-adjusted estimator. Same center, ~25% tighter.
- $SD(\text{actual}) = 0.012$  (was 0.016 without adjustment)
- Mean HC2 SE = 0.012 (was 0.016 without adjustment)

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- “Block on what you can, adjust for what you can’t.”

# Blocking: the estimator

Within each block  $j$ , compute the block-specific diff-in-means  $\hat{\tau}_j$ . Then weight by block size:

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If treatment probability is not equal in all blocks, the Lin-style interacted regression, IPW, and the block-DIM average all recover the ATE and are numerically identical (Shinkre & Hazlett 2025).

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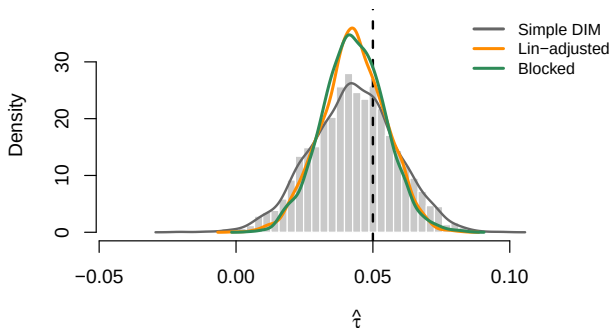
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## Add blocking on past-turnout quartiles



- Green = dist. of  $\hat{\tau}$  from blocking
- Matches Lin here, not always the case.
- SD(actual) = 0.012; Mean HC2 SE = 0.012

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- Formal variance comparison in the appendix.

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- **Result:** the “neighbors” treatment moved turnout by  $\approx 8$  percentage points — enormous for a piece of mail.
- Relevance here:
  - The blocking is powerful – shows importance of good design.
  - Randomization is clustered at household level – we will talk about its implications next.

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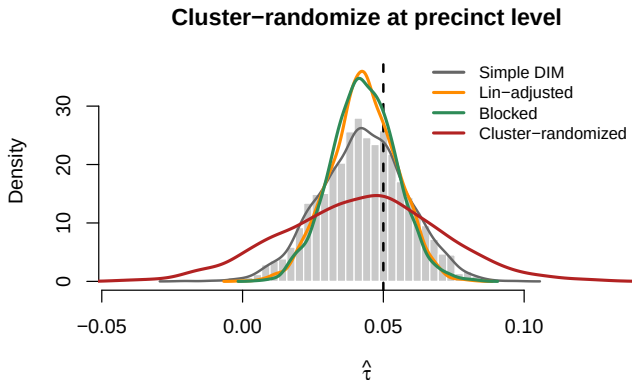
- 1 **Clustered assignment/ cluster-randomization.** Sometimes necessary to assign treatment at group level (precinct, village, school). This is bad news for actual variance/noise and may mess up estimated SE relative to that.
- 2 **Dependent residuals within clusters.** Even with individual assignment, residuals can be correlated within a group (e.g. panel data); estimated SEs that ignore this can be wrong, usually too small.

### Cluster-randomize at precinct level



# Simulation: cluster-randomize at the precinct level

Same DGP but now everyone in a precinct gets the same  $D$ .



Eek! Red curve shows you get much bigger gaps (high variance)

# And the (HC2) SE is wrong...but CR-SE fixes it.

| Estimator                      | SD(actual)   | HC2 SE       | CR-SE (precinct) |
|--------------------------------|--------------|--------------|------------------|
| Simple DIM ( $\ln(Y \sim D)$ ) | 0.016        | 0.016        | 0.016            |
| Lin-adjusted                   | 0.012        | 0.012        | —                |
| Blocked (block FEs)            | 0.012        | 0.012        | —                |
| Cluster-randomized             | <b>0.028</b> | <b>0.016</b> | <b>0.028</b>     |

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- HC2 assumes independent residuals but cluster-influences + cluster-assignment badly violates this.
- Cluster-robust SE (clustered at the precinct level) recovers the actual sampling variance almost exactly. In R:

```
sandwich::vcovCL(fit, cluster = precinct_id).
```

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**Both at once is the dangerous case.** Cluster-level influences on  $Y$  and cluster-assignment – our case above.

# Where “both at once” shows up in practice

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- **Treatment probabilities that differ by cluster.** Even with individual assignment, asymmetric  $T/C$  proportions correlated with  $u_c$  prevent it from cancelling cleanly.
- **Heterogeneous treatment effects by cluster.**  $\tau_c$  varies across clusters; the residual carries a  $\tau_c D_i$  piece that is cluster-correlated on the treated side and doesn't cancel.

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# Catches with CR-SEs

- **They are not free.** CR-SEs can be conservative when— intervals widen, power drops.
- **But it is not** as bad as “having only  $N_{cluster}$  observations”: CR-SE estimates within-cluster correlation.
- **Few clusters is a real problem.** The standard CR-SE estimator (`vcovCL`) is downward biased with fewer than (roughly!) 50 clusters.
- **Fixes for few clusters:**
  - “Block bootstrap”: Resample whole clusters with replacement.
  - Wild cluster bootstrap `fwildclusterboot` in R.
  - Randomization inference at the cluster level.
  - Aggregate to the cluster level and run the analysis there (extremely conservative)

# Takeaways

- **Unbiasedness is not enough.** The same experiment can be run more or less precisely; that's what we're paying attention to today.
- **Adjustment is good.** Covariate adjustment recovers real precision when  $X$  predicts  $Y$ . Use Lin's interacted form, with HC2 SEs; pre-specify the covariates.
- **Design is paramount.** Block on what you can; adjust for what you can't. The two are complements, not substitutes.
- **Cluster on the source of dependence in your residuals**, not on every group you can think of. Watch out for too-few-clusters and use an appropriate alternative when needed.
- Next: what if we can't randomize at all? **Selection on observables.**

# Appendix

# Frisch-Waugh-Lovell (FWL) theorem

Two facts:

- 1 Any bivariate regression of  $Y$  on  $X$  has  $\hat{\beta} = \widehat{\text{Cov}}(Y, X) / \widehat{\text{Var}}(X)$ .
- 2 The coefficient on the  $k$ th regressor in a multivariate regression can be obtained by (i) regressing  $X_k$  on the other  $X$ 's and saving the residual  $\tilde{X}_k$ , then (ii) regressing  $Y$  (or its residual  $\tilde{Y}$ ) on  $\tilde{X}_k$ .

Together:

$$\hat{\beta}_k = \frac{\widehat{\text{Cov}}(\tilde{Y}_i, \tilde{X}_{ki})}{\widehat{\text{Var}}(\tilde{X}_{ki})}$$

# Why experimental findings are robust to specification

Apply FWL to  $D$ :

$$\hat{\beta}_D = \frac{\widehat{\text{Cov}}(\tilde{Y}_i, \tilde{D}_i)}{\widehat{\text{Var}}(\tilde{D}_i)}$$

where  $\tilde{D}_i$  is the residual from regressing  $D_i$  on  $X_i$ .

- Under randomization,  $D \perp\!\!\!\perp X$ , so on average  $\tilde{D} \approx D$ .
- Hence adjusting for  $X$  doesn't move  $\hat{\beta}_D$  much in expectation — it just removes the variance contribution from chance imbalances.
- This is why experimental results are typically robust to “what did you control for?”

# Lin (2013): why the demeaning?

$$Y_i = \alpha + \tau_{Lin} D_i + \beta'_1 (X_i - \bar{X}) + \beta'_2 D_i (X_i - \bar{X}) + \varepsilon_i$$

- $\mathbb{E}[Y_i \mid D_i = 0, X_i = \bar{X}] = \alpha$
- $\mathbb{E}[Y_i \mid D_i = 1, X_i = \bar{X}] = \alpha + \tau_{Lin}$
- So  $\tau_{Lin}$  is the comparison of treated vs. control *at the mean of  $X$* .
- Because  $\mathbb{E}_X[\text{ATE}(X)] = \tau_{Lin}$  for this model, it recovers the ATE without assuming constant treatment effects.
- Plain OLS without the interaction works in large samples but is biased  $O(1/N)$  (Freedman 2008). Lin's form removes this and is now standard practice.

# When does blocking help? (variance comparison)

Models for complete and block-randomized designs:

$$Y_i = \alpha + \tau_{CR} D_i + \epsilon_i$$

$$Y_i = \alpha + \tau_{BR} D_i + \sum_{j=2}^J \beta_j B_{j[i]} + \epsilon_i^*$$

Then

$$\text{Var}[\hat{\tau}_{CR}] = \frac{\sigma_\epsilon^2}{\sum_i (D_i - \bar{D})^2}$$

$$\text{Var}[\hat{\tau}_{BR}] = \frac{\sigma_{\epsilon^*}^2}{\sum_i (D_i - \bar{D})^2 (1 - R_D^2)}$$

where  $R_D^2$  is from regressing  $D$  on the block dummies.

# When does blocking help? (intuition)

- If treatment proportions are the same in every block,  $R_D^2 = 0$  and the denominator is unchanged — blocking doesn't *cost* you anything.
- If the blocks strongly explain  $Y$ , then  $\sigma_{\varepsilon^*}^2 \ll \sigma_{\varepsilon}^2$  — blocking *wins* a lot.
- Combine: blocking on a strong predictor of  $Y$ , with equal-share randomization, weakly improves precision and usually does so a lot.

# Cluster-robust variance: a sketch

For OLS  $\hat{\beta} = (X'X)^{-1}X'Y$ , the cluster-robust variance is

$$\widehat{\text{Var}}_{CR}(\hat{\beta}) = (X'X)^{-1} \left( \sum_{c=1}^C X'_c \hat{e}_c \hat{e}_c' X_c \right) (X'X)^{-1}$$

- $X_c$  and  $\hat{e}_c$  are the design matrix and residuals for cluster  $c$ .
- This collapses cluster-level scores, so it allows arbitrary intra-cluster correlation in residuals while assuming clusters are independent.
- The “effective  $N = C$ ” approximation is the limit case  $\rho \rightarrow 1$ . The actual penalty is data-driven via  $\hat{e}_c$ .
- With  $C \lesssim 30$ , this estimator is downward biased; use a wild cluster bootstrap (Cameron, Gelbach, & Miller 2008) instead.

# LPM caveats (binary outcomes)

Using OLS / LPM with a binary  $Y$ :

- **Pro:** unbiased for the ATE under random assignment; coefficients are interpretable as probability differences; works in every standard software environment.
- **Cons:**
  - Predicted probabilities can fall outside  $[0, 1]$ .
  - Heteroskedasticity is built in — use robust SEs.
  - In observational settings with extreme covariates, linearity can be a poor approximation.
- For the experimental settings in this lecture, none of these are reasons to abandon LPM.